

DAC Domain — On-Chain .dac Identity Registry

Explorer-Friendly Event Layer — v0.5 Archived Prototype → v1.0.0

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Environment	GitHub Pages · WSL Ubuntu · Local DAC Node RPC (:8546) · Indonesia / CGNAT
Report Type	Application Development & Infrastructure Validation — On-Chain Identity Registry, Local RPC Deployment, and Event Readability Testing
Version Scope	DAC Domain v0.5 Archived Prototype → v1.0.0 Explorer-Friendly Registry
Relates to	Critical RPC Availability Issue & Network Access Bottleneck (05 May 2026) · Testnet Infrastructure Validation Report (07 May 2026) · EIP-1559 Compatibility & Txpool Analysis (09 May 2026) · Frontend–Blockchain State Desynchronization After Successful NFT Mint (10 May 2026) · DAC SENDER — Testnet Transaction Interface (22 May 2026)
Report Archive	https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/Testnet (Inception) Technical Reports

Executive Summary

This report documents the design, deployment, validation, and iteration of DAC Domain — a community-built experimental on-chain .dac identity registry for DAC Testnet. The project began as a frontend and smart-contract prototype and evolved into a deployed v1 explorer-friendly registry with readable event data, reproducible deployment scripts, and a GitHub-hosted static dApp.

Testing was performed from a self-hosted DAC node through local RPC endpoint `http://127.0.0.1:8546`, because the official public RPC had shown instability during previous infrastructure investigations and during this project's deployment preparation. The result is a working .dac registry with registered test domain `jeruzsalem.dac`, primary-domain selection, and decoded event logs that expose the domain name as readable data.

Finding	Status	Detail
DAC Domain v1 deployed	Confirmed	Registry v1 deployed at <code>0x90F07E7E...a090F</code> on DAC Testnet
Frontend dApp operational	Confirmed	GitHub-hosted static interface with wallet connection, domain registration, and Domain Management
Self-hosted local RPC deployment	Confirmed	Deployment and validation performed through local DAC node RPC <code>:8546</code>
v0.5 full end-to-end prototype	Archived Prototype	Functional registry, registration, and primary-domain flow completed — event schema archived
v1 readable domain events	Confirmed	Decoded <code>DomainRegistered</code> and <code>PrimaryDomainSet</code> logs include name: <code>jeruzsalem.dac</code>

Explorer-wide address replacement	Future integration	Requires explorer/indexer-side name service integration; not automatic from the registry alone
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Tool Overview

DAC Domain is a no-backend, static web application and smart contract registry built to test human-readable wallet identity on DAC Testnet. The frontend connects to a browser wallet, checks .dac domain availability, submits registration transactions, loads owned domains, and allows users to select a primary domain.

The registry is intentionally experimental. It does not claim to be an official DAC name service. Its main value is to provide a testnet identity layer and a measurable on-chain data source for future explorer, indexer, or reporting workflows.

Component	Description
Frontend	Static GitHub Pages dApp with DAC-themed UI, rotating background, wallet connection, availability checking, registration duration selection, and Domain Management.
Smart Contract	DACDomainRegistry stores .dac records, owner-to-domain mappings, expiry data, target wallet data, primary domain state, and event logs.
Deployment Layer	Hardhat project configured for DAC Testnet with local RPC support, manual gas deployment, and nonce-based replacement tooling.
Observation Layer	CLI scripts for domain resolution and event decoding provide reproducible technical evidence for reports.

Live Interface & Repository Links

Resource	Location
Repository Path	EdLWEISS186/dac-dual-node-cgnat-setup / DAC-Domain/
Live Frontend	https://edlweiss186.github.io/dac-dual-node-cgnat-setup/DAC-Domain/
DAC Explorer	https://exptest.dachain.tech
Public RPC	https://rpctest.dachain.tech/
Local RPC Used	http://127.0.0.1:8546

Testing Environment

Network	DAC Testnet
Chain ID	21894 / 0x5586

Native Currency	DACC
TLD	.dac
Local RPC Used	http://127.0.0.1:8546
Public RPC	https://rpctest.dachain.tech/
Explorer	https://exptest.dachain.tech
Deployer / Primary Wallet	0x870ad63acc507cdfd878F170606d19ae78988AFE
Node.js / npm	Node.js v24.14.1 · npm 11.11.0
Hardhat	2.28.6
Solidity	0.8.20, optimizer enabled, 200 runs
Frontend Runtime	Static HTML/CSS/JavaScript + ethers.js in browser

Local RPC Validation

```
eth_chainId    → {"jsonrpc":"2.0","id":1,"result":"0x5586"}
eth_blockNumber → {"jsonrpc":"2.0","id":1,"result":"0xe444b5"}

Deployer : 0x870ad63acc507cdfd878F170606d19ae78988AFE
Balance  : 12.264253556993023242 DACC
```

The local RPC returned the correct chain ID (0x5586 / 21894), confirming that deployment and contract reads were performed against DAC Testnet.

Architecture

The DAC Domain architecture follows a lightweight dApp pattern: a static frontend, browser wallet provider, JSON-RPC access layer, and an on-chain registry contract. This keeps the application simple to host while still producing persistent on-chain identity data.

```
End User
  | Connect wallet / sign tx / read state
  ▼
DAC Domain Frontend
  ├── index.html
  ├── src/app.js
  ├── src/style.css
  ├── assets/background/bg-1.png ... bg-3.png
  | ethers.js
  ▼
Browser Wallet Provider
  | JSON-RPC
  ▼
Local DAC RPC Node (http://127.0.0.1:8546)
  ▼
DAC Testnet
  ├── DACDomainRegistry v1
  |   0x90F07E7EFa772c40B90d68BB54267Ea0658a090F
  ▼
Explorer / Reporting Layer
  ├── tx inspection
  ├── event/log inspection
  └── future indexer/name-service integration
```

Layer	Responsibility
Frontend	Provides human-readable .dac registration and management interface.
Wallet Provider	Signs registration, renewal, and primary-domain transactions.
RPC Layer	Sends reads and writes to DAC Testnet. Local RPC was preferred for deployment reliability.
Registry Contract	Stores domain ownership, target wallet, expiry, owned-domain list, and primary domain state.
Event Layer	Emits registration and primary-domain events for reporting and potential indexing.
Explorer / Indexer	Can inspect transactions and logs. Explorer-wide address replacement requires additional integration.

Version Evolution

DAC Domain was intentionally developed in two distinct phases. v0.5 proved the full application flow. v1 preserved the working behavior while improving event readability for explorer-facing data and technical reporting.

Version	Purpose	Result
v0.5 Archived Prototype	Prove the .dac registry concept end-to-end: UI, deployment, registration, resolve, Domain Management, and primary-domain flow.	Functional success. Archived because event schema was not ideal for readable explorer logs.
v1.0.0 Explorer-Friendly Registry	Improve event design by emitting indexed nameHash plus readable string name while preserving the registry workflow.	Deployed and validated. DomainRegistered and PrimaryDomainSet decode with name: jeruzsalem.dac.

Frontend UI/UX Implementation

The UI/UX phase converted the registry concept into a usable testnet application. The final interface kept a distinct DAC infrastructure identity with a dark industrial visual style.

Feature	Implementation
Visual Identity	DAC-themed rotating background, 10-second fade transition, 10% dark overlay, and glass readability panels.
Wallet Area	Single wallet box showing Connect Wallet before connection and Balance DACC Short Address after.
Register Domain	Input normalizes to .dac, removes invalid characters, checks availability, and shows duration pricing only when relevant.
Pricing	1 Year = 5 DACC · 2 Years = 8 DACC · 3 Years = 10 DACC

Domain Management	Displays owned domains, target wallet, expiry date, primary-domain state, renewal action, and empty-state messaging.
Advanced Settings	Developer-only registry address fallback separated from the main user flow.

Smart Contract Scope

The DACDomainRegistry contract provides a custom .dac registry model. It is not intended to replace official explorer behavior by itself, but provides a working on-chain source of truth for domain records.

Functionality	Status
.dac name validation	Confirmed
1/2/3 year registration pricing	Confirmed
Target wallet assignment	Confirmed
Expiry timestamp tracking	Confirmed
Domain renewal	Implemented
Owned-domain listing	Confirmed
Primary-domain selection	Confirmed
Primary-domain clearing	Implemented
Readable v1 event data	Confirmed
Contract owner withdrawal	Implemented

Technical Findings

Finding 1 — Local RPC Was the Reliable Deployment Path

The official public RPC was not used as the primary deployment path because prior tests and this deployment preparation showed instability. The self-hosted local node RPC `http://127.0.0.1:8546` provided a reliable path for contract deployment, verification, domain registration, and event decoding.

RPC Path	Role	Observed Result
Public RPC	General access	Known instability — unsuitable as primary deployment path during this workflow
Local RPC :8546	Self-hosted node	Validated chain ID, balance, deployer, deployment, registration, and reads

Finding 2 — v0.5 Deployment Required Pending TX Replacement

The initial v0.5 contract creation transaction became pending because of a low gas price. A replacement deployment was submitted using the same nonce (2627) and a higher gas price (100 gwei). Reusing the same nonce preserved the deterministic contract creation address.

```
GAS_PRICE_GWEI=100 REPLACE_NONCE=2627 \  
npx hardhat run scripts/replace-deploy.js --network dacTestnet  
  
Receipt status : 1  
Block          : 14959901  
Contract       : 0x72BD75723ADA5e37F6bA4b8909864c3bbaBccB63
```

This demonstrated a practical testnet deployment recovery workflow: detect pending state, confirm nonce, replace with higher gas, preserve the expected contract address, and write deployment metadata immediately after confirmation.

Finding 3 — v0.5 Worked Functionally but Was Not Final

v0.5 proved the end-to-end domain registry workflow. The UI detected jeruzsalem.dac as available, registered it, displayed it in Domain Management, and allowed it to be set as the primary domain. The resolve script confirmed owner, target, expiry, registration year, and active status.

```
Resolving .dac domain...  
Domain           : jeruzsalem.dac  
Registry         : 0x72BD75723ADA5e37F6bA4b8909864c3bbaBccB63  
Owner            : 0x870ad63acc507cdfd878F170606d19ae78988AFE  
Target           : 0x870ad63acc507cdfd878F170606d19ae78988AFE  
Registered At    : 2026-06-05T15:02:17.000Z  
Expires At       : 2027-06-05T15:02:17.000Z  
Registration Years: 1  
Active           : true
```

The limitation was not functional correctness — it was event readability. v0.5 used string indexed name, which makes dynamic strings appear as hashes in event topics rather than readable domain text.

Finding 4 — v1 Improved Event Readability

v1 replaced the less-readable indexed string pattern with a filterable hash plus readable string data.

```
v0.5 event (limitation):  
  string indexed name  
  
v1 event (improved):  
  bytes32 indexed nameHash  
  string  name
```

This makes v1 more suitable for reporting, event decoding, and potential future indexing. The name can be read as plain text while the hash remains useful for filtering.

Finding 5 — Registry Data and Explorer Address Replacement Are Different Layers

A successful .dac registration does not automatically make DAC Explorer display from jeruzsalem.dac instead of the raw wallet address for every normal wallet transaction. Normal native transfers do not call the DAC Domain contract, and the explorer does not automatically know that this custom registry should be used as a name service.

Capability	v1 Status
Resolve jeruzsalem.dac to wallet address	Confirmed

Show readable domain name in decoded registry events	Confirmed
Set primary domain on registry	Confirmed
Automatically replace all explorer addresses with .dac names	Requires explorer/indexer integration

v1 Deployment and Validation Evidence

Pre-Deployment Nonce Check

```

Wallet : 0x870ad63acc507cdfd878F170606d19ae78988AFE
Balance : 8.145905668985909878 DACC
Latest nonce : 2631
Pending nonce : 2631
RPC gasPrice : 42688

```

Both latest and pending nonce were 2631, confirming no unresolved pending transaction before deployment.

Deployment Output

```

Deploying DACDomainRegistry v1...
Network : dacTestnet
Chain ID : 21894
Deployer : 0x870ad63acc507cdfd878F170606d19ae78988AFE
Balance : 8.145905668985909878 DACC
Gas price : 100 gwei Gas limit : 2500000

Deployment TX : 0x4e3ab77b099bb7e2149f17dc9465e5e760c0d3dd96790df11aafce482eeb1c90
Contract : 0x90F07E7EFa772c40B90d68BB54267Ea0658a090F
Status : 1 Block : 14960256

```

Contract Read Validation

```

Version : v1.0.0-explorer-friendly
Contract : 0x90F07E7EFa772c40B90d68BB54267Ea0658a090F
Code exists : true
TLD : .dac
Price 1 year : 5.0 DACC
Price 2 years : 8.0 DACC
Price 3 years : 10.0 DACC
NameHash jeruzsalem.dac :
0x25992959442c572a45da6926192bd643f9c52487d3bc370bfa8fb24a5f52f246

```

Domain Registration and Primary Domain Transactions

Action	TX Hash	Result
Register jeruzsalem.dac	0xb8524d29a47261be3ea65eefe9fe089a11d4a806d8e9b3ace5359c9ad9e110ce	Success
Set primary domain	0x0c093b06cfb5525d4734bb5af2539a970f6ab1b280cdbb763a6728d9a1fdf82a	Success

Local Node Transaction Logs

```
INFO [06-05|23:26:03.296] Submitted transaction
    hash=0xb8524d29a47261be3ea65eefe9fe089a11d4a806d8e9b3ace5359c9ad9e110ce
    from=0x870ad63acc507cdfd878F170606d19ae78988AFE nonce=2632
    recipient=0x90F07E7EFa772c40B90d68BB54267Ea0658a090F value=5,000,000,000,000,000

INFO [06-05|23:27:01.094] Submitted transaction
    hash=0x0c093b06cfb5525d4734bb5af2539a970f6ab1b280cdbb763a6728d9a1fdf82a
    from=0x870ad63acc507cdfd878F170606d19ae78988AFE nonce=2633
    recipient=0x90F07E7EFa772c40B90d68BB54267Ea0658a090F value=0
```

Resolve Script Validation

```
Domain      : jeruzsalem.dac
Registry    : 0x90F07E7EFa772c40B90d68BB54267Ea0658a090F
Owner       : 0x870ad63acc507cdfd878F170606d19ae78988AFE
Target      : 0x870ad63acc507cdfd878F170606d19ae78988AFE
Registered  : 2026-06-05T16:26:17.000Z
Expires     : 2027-06-05T16:26:17.000Z
Years       : 1
Active      : true
```

Event Decode Validation

A reusable event decoder script was added to make log validation reproducible. The script reads transaction receipts, parses logs using the compiled ABI, and prints decoded event fields.

```
npm run decode-events:dac -- \
  0xb8524d29a47261be3ea65eefe9fe089a11d4a806d8e9b3ace5359c9ad9e110ce \
  0x0c093b06cfb5525d4734bb5af2539a970f6ab1b280cdbb763a6728d9a1fdf82a
```

Decoded DomainRegistered Event

```
TX      : 0xb8524d29a47261be3ea65eefe9fe089a11d4a806d8e9b3ace5359c9ad9e110ce
Status  : 1   Block : 14960369

Event   : DomainRegistered
  nameHash : 0x25992959442c572a45da6926192bd643f9c52487d3bc370bfa8fb24a5f52f246
  name     : jeruzsalem.dac
  owner    : 0x870ad63acc507cdfd878F170606d19ae78988AFE
  target   : 0x870ad63acc507cdfd878F170606d19ae78988AFE
  registrationYears : 1
  expiresAt : 1812212777
  pricePaid : 5000000000000000000
  timestamp : 1780676777
```

Decoded PrimaryDomainSet Event

```
TX      : 0x0c093b06cfb5525d4734bb5af2539a970f6ab1b280cdbb763a6728d9a1fdf82a
Status  : 1   Block : 14960373

Event   : PrimaryDomainSet
  owner    : 0x870ad63acc507cdfd878F170606d19ae78988AFE
  nameHash : 0x25992959442c572a45da6926192bd643f9c52487d3bc370bfa8fb24a5f52f246
  name     : jeruzsalem.dac
  timestamp : 1780676837
```

This is the key v1 validation result. The domain name is no longer only hidden behind an indexed dynamic topic. It is present as readable event data — name: jeruzsalem.dac — in both the DomainRegistered and PrimaryDomainSet events.

Deployment Snapshots

v0.5 Archived Prototype

Registry	0x72BD75723ADA5e37F6bA4b8909864c3bbaBccB63
Successful Deploy TX	0xe97e92b860257dacdbb791ad5d12862127d9e8d0b74c28e063df44f2f68968e5
Replaced TX	0x0514ff8cf8949c4885f34da793f4a2ada8b10b3dc02c15be80bd3b981b992af6
Replacement Nonce	2627
Gas Price	100 gwei
Block	14959901
Registered Domain	jeruzzalem.dac
Status	Archived Prototype

v1.0.0 Explorer-Friendly Registry

Registry	0x90F07E7EFa772c40B90d68BB54267Ea0658a090F
Deployment TX	0x4e3ab77b099bb7e2149f17dc9465e5e760c0d3dd96790df11aafce482eeb1c90
Deployment Block	14960256
Deployment Nonce	2631
Registration TX	0xb8524d29a47261be3ea65eefe9fe089a11d4a806d8e9b3ace5359c9ad9e110ce
Primary Domain TX	0x0c093b06cfb5525d4734bb5af2539a970f6ab1b280cdbb763a6728d9a1fdf82a
NameHash	0x25992959442c572a45da6926192bd643f9c52487d3bc370bfa8fb24a5f52f246
Status	Deployed and validated

Repository and Reproducibility

The final project files were committed and pushed to GitHub after syncing with remote README updates. The initial push was rejected because the remote branch contained newer web edits; this was resolved through a safe rebase without force pushing.

```
git fetch origin main
git pull --rebase origin main
git push origin main
```

```
Final commit: a75b0bc Add DAC Domain v1 explorer-friendly registry
```

File / Path	Purpose
DAC-Domain/contracts/DACDomainRegistry.sol	Active v1 registry contract source.
DAC-Domain/index.html	Static frontend entry point.
DAC-Domain/src/app.js	Wallet, registry interaction, domain checking, registration, and management logic.
DAC-Domain/src/style.css	DAC-themed UI styling and readability layers.
DAC-Domain/scripts/deploy-v1.js	Manual-gas v1 deployment script.
DAC-Domain/scripts/replace-deploy.js	v0.5 pending deployment replacement script.
DAC-Domain/scripts/register.js	CLI registration helper.
DAC-Domain/scripts/resolve.js	CLI resolver / record inspection helper.
DAC-Domain/scripts/decode-events.js	Reproducible event decoding helper.
DAC-Domain/deployments/dac-testnet.json	Active v1 deployment metadata.
DAC-Domain/docs/archive/	v0.5 and v1 archive notes for report source material.

Contribution Value

DAC Domain extends the existing DAC testnet contribution series from infrastructure diagnostics into identity tooling. It contributes a practical, deployed example of how a community-built registry can be used to observe wallet identity, frontend-contract interaction, and event readability behavior on DAC Testnet.

Contribution Area	Value
Application Layer	A usable static dApp for .dac registration and management.
Smart Contract Layer	A custom registry contract with pricing, expiry, ownership, target resolution, and primary-domain behavior.
Infrastructure Layer	Deployment and validation through self-hosted local RPC, reducing dependency on unstable public RPC.
Observability Layer	Decoded event logs provide readable evidence for reporting and potential future indexer work.
Documentation Layer	Archive notes, README, scripts, and deployment metadata make the workflow reproducible.

Recommendations

For DAC Domain

1. Publish or preserve ABI and deployment metadata alongside the frontend for easier third-party contract reads.
2. Add an optional reverse lookup helper that returns the primary domain for a wallet address.

3. Create an event snapshot script that exports decoded registry events into JSON/Markdown for reports.
4. Add a small frontend indicator explaining that explorer-wide address replacement requires indexer support.
5. Consider a lightweight off-chain indexer prototype that maps wallet address to primary .dac domain from contract events.

For DAC Testnet Infrastructure / Explorer Integration

1. Provide an official path for testnet name-service integration or custom address labels.
2. Expose documented explorer/indexer hooks for community-built registries.
3. Stabilize public RPC access so deployment and frontend reads do not depend only on local nodes.
4. Document recommended wallet/RPC/gas settings for contract deployments on DAC Testnet.

Conclusion

DAC Domain v1 is a deployed and validated on-chain .dac identity registry for DAC Testnet. v0.5 proved the functional path. v1 improved the project's reporting and observability value by emitting readable domain names in decoded event logs.

The project does not automatically make DAC Explorer replace every wallet address with .dac names, because that behavior belongs to the explorer/indexer layer. However, the registry now provides the necessary on-chain data and readable event structure for that future integration path.

As a community-built testnet contribution, DAC Domain is valuable not only as a frontend application but also as a practical infrastructure experiment: it combines local RPC deployment, smart contract iteration, frontend state management, event design, and reproducible validation into a single testnet reportable workflow.

Layer / Component	Final Status
Frontend UI/UX	Operational — GitHub-hosted static dApp
v0.5 Registry	Archived — functional prototype
v1 Registry	Deployed and validated
Local RPC Deployment	Confirmed — self-hosted DAC node :8546
Domain Registration	Confirmed — jeruzzalem.dac
Primary Domain	Confirmed — jeruzzalem.dac
Readable Event Logs	Confirmed — DomainRegistered and PrimaryDomainSet
Explorer-wide name replace	Future integration — requires explorer/indexer support

Supporting Evidence — Key Output Reference

The following extracts are compact evidence references for this report. Full archive notes and repository artifacts are maintained under [DAC-Domain/docs/archive/](#).

```
Active v1 Registry : 0x90F07E7EFa772c40B90d68BB54267Ea0658a090F
Registered Domain : jeruzzalem.dac
```

Deployment TX	: 0x4e3ab77b099bb7e2149f17dc9465e5e760c0d3dd96790df11aafce482eeb1c90
Registration TX	: 0xb8524d29a47261be3ea65eefe9fe089a11d4a806d8e9b3ace5359c9ad9e110ce
Primary Domain TX	: 0x0c093b06cfb5525d4734bb5af2539a970f6ab1b280cdbb763a6728d9a1fdf82a
Final Commit	: a75b0bc Add DAC Domain v1 explorer-friendly registry